

Total No. of Questions : 8]

SEAT No. :

P190

[Total No. of Pages : 3

[5871]-712

B.E. (Computer Engineering)

MACHINE LEARNING

(2015 Pattern) (Semester - II)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Assume suitable data if necessary.
- 3) Neat diagrams must be drawn wherever necessary.
- 4) Figures to the right indicates full marks.

- Q1) a) What is Overfitting and Underfitting in machine learning model? Explain with an example. [6]
- b) Differentiate with an example Label encoder and One Hot encoder for managing categorical data. [6]
- c) How Ridge Regression help for regularizing linear models? Write Scikit-Learn code for Ridge Regression. [8]

OR

- Q2) a) Explain Data formats for supervised learning problem with example. [6]
- b) Write a Scikit-learn code to manage missing values of the given 3-D dataset using mean and median strategy : [6]

X	V	Z
1	NaN	2
2	3	NaN
1	4	2

- c) What do you mean by Logistic Regression? Explain with example. [8]
- Q3) a) What do you mean by Support Vector Machine? Explain with example. [8]
- b) Write short notes on : [9]
- i) Bernoulli naive Bayes
 - ii) Multinomial naive Bayes
 - iii) Gaussian naive Bayes

P.T.O.

OR

Q4) a) Explain with example how Bayes theorem helps to classify the data using Naïve Bayes Classification technique. [8]

b) Differentiate between linear SVM and non-linear SVM. [4]

c) Write and Explain Scikit-learn code for Gaussian naive Bayes classifier. [5]

Q5) a) Explain with example the role of attribute selection measure in building decision tree. [8]

b) Write and explain data formats for clustering technique. [5]

c) Explain Elbow method for finding optimal number of clusters. [4]

OR

Q6) a) Consider the following data set consisting of the scores of two variables on each of seven individuals: Apply Kmeans clustering to cluster this data into 2 clusters. [8]

Subject	A	B
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	5.0	7.0
5	3.5	5.0
6	4.5	5.0
7	3.5	4.5

b) Write short notes on : [9]

i) Bagging

ii) Boosting

iii) Random Forests

Q7) a) Explain with example Naïve User based Recommendation systems. [8]

b) What is agglomerative clustering? Explain with example. [8]

OR

- Q8) a)** Perform the complete link hierarchical clustering using similarity matrix given below : Show the Dendrograms **[8]**

	A	B	C	D
A	0	1	4	5
B		0	2	6
C			0	3
D				0

- b) Explain with example common architectural principles of deep networks. **[8]**

